CHAPTER 13

STATISTICS

CHAPTER ESSENCE AT A GLANCE:

Statistics measures how scattered your data is! While Mean tells you where the center is, Variance (σ^2) and Standard Deviation (σ) tell you how widely spread the values are around that center. A lower Coefficient of Variation (CV) means higher stability and consistency!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Measures of Dispersion:** Dispersion quantifies how data values scatter around the central average. Range, Mean Deviation, Variance, and Standard Deviation.

• **Mean Deviation (MD):** MD about Mean: $\Sigma |x_i - x|$ over N. MD about Median: $\Sigma |x_i - M|$ over N. Always take absolute positive differences $|x_i - A|$!

• **Variance σ^2 & Standard Deviation σ:** Ungrouped variance $\sigma^2 = [\Sigma x_i^2$ over N] - $(x)^2$. Standard Deviation σ = sqrt(Variance).

• **Grouped Variance Formula:** $\sigma^2 = [$ (1 over N) $\Sigma f_i x_i^2] - (x)^2$, where N = Σf_i and x = (1 over N) $\Sigma f_i x_i$.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\bar{x} = \frac{\sum f_i x_i}{N} \quad (N = \sum f_i)$$

Arithmetic Mean of Grouped Frequency Distribution

$$\text{MD}(\bar{x}) = \frac{\sum f_i |x_i - \bar{x}|}{N}, \quad \text{MD}(M) = \frac{\sum f_i |x_i - M|}{N}$$

Mean Deviation about Mean and Median

$$\sigma^2 = \frac{1}{N}\sum f_i (x_i - \bar{x})^2 = \frac{1}{N}\sum f_i x_i^2 - \bar{x}^2$$

Variance of Grouped Data (Discrete Frequency Form)

$$\sigma = \sqrt{\sigma^2}, \quad \text{CV} = \frac{\sigma}{\bar{x}} \times 100 \quad (\text{Coefficient of Variation})$$

Standard Deviation and Measure of Relative Dispersion

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Between two teams or series, the one with the SMALLER Coefficient of Variation (CV) is MORE CONSISTENT, MORE STABLE, and LESS VARIABLE!

• **Alert:** If every observation in a dataset is multiplied by constant k, the new standard deviation is simply k × σ!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Find the variance of the first n natural numbers. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formulas for Sums]** Mean x_{bar} = (n + 1) over 2. Σx_i^2 = n(n + 1)(2n + 1) over 6. [1 Mark]

• **[Step 2: Variance]** sigma^2 = (1 over n) Σx_i^2 - $(x_{bar})^2$ = (n + 1)(2n + 1) over 6 - $(n + 1)^2$ over 4 = [(n + 1) over 12] [2(2n + 1) - 3(n + 1)] = [(n + 1) over 12] (4n + 2 - 3n - 3) = (n^2 - 1) over 12. [1 Mark]

• **[Boxed Result: Variance = (n^2 - 1) over 12]**

2. If every observation in a dataset is increased by 5, what happens to the mean and standard deviation? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Effect on Mean]** If each $x_i \to x_i + 5$, the new mean increases by 5: $x_{bar}' = x_{bar} + 5$. [1 Mark]

• **[Step 2: Effect on Standard Deviation]** Since deviations from the mean $(x_i' - x_{bar}') = (x_i + 5) - (x_{bar} + 5) = x_i - x_{bar}$ remain identical, the standard deviation is UNCHANGED (σ' = σ). [1 Mark]

• **[Boxed Result: Mean increases by 5; Standard deviation remains unchanged]**

3. Find the mean deviation about the mean for the data: 4, 7, 8, 9, 10, 12, 13, 17. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Calculate Mean]** Mean x_{bar} = (4 + 7 + 8 + 9 + 10 + 12 + 13 + 17) over 8 = 80 over 8 = 10. [1 Mark]

• **[Step 2: Absolute Deviations $|x_i - x_{bar}|$]** |4 - 10| = 6; |7 - 10| = 3; |8 - 10| = 2; |9 - 10| = 1; |10 - 10| = 0; |12 - 10| = 2; |13 - 10| = 3; |17 - 10| = 7. Sum of $|d_i|$ = 6 + 3 + 2 + 1 + 0 + 2 + 3 + 7 = 24. [1 Mark]

• **[Step 3: Mean Deviation]** MD(x_{bar}) = (1 over n) $\Sigma |x_i - x_{bar}|$ = 24 over 8 = 3. [1 Mark]

• **[Boxed Result: Mean Deviation about Mean = 3]**

4. Find the mean deviation about the median for the data: 13, 17, 16, 14, 11, 13, 10, 16, 11, 18, 12, 17. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Arrange Data in Ascending Order]** 10, 11, 11, 12, 13, 13, 14, 16, 16, 17, 17, 18 (n = 12, even). Median M = [6th term + 7th term] over 2 = (13 + 14) over 2 = 13.5. [1 Mark]

• **[Step 2: Absolute Deviations $|x_i - M|$]** |10 - 13.5| = 3.5; |11 - 13.5| = 2.5 (twice = 5.0); |12 - 13.5| = 1.5; |13 - 13.5| = 0.5 (twice = 1.0); |14 - 13.5| = 0.5; |16 - 13.5| = 2.5 (twice = 5.0); |17 - 13.5| = 3.5 (twice = 7.0); |18 - 13.5| = 4.5. Sum = 3.5 + 5.0 + 1.5 + 1.0 + 0.5 + 5.0 + 7.0 + 4.5 = 28. [1 Mark]

• **[Step 3: Mean Deviation]** MD(Median) = 28 over 12 = 7 over 3 ≈ 2.33. [1 Mark]

• **[Boxed Result: Mean Deviation about Median = 7 over 3 ≈ 2.33]**

5. Find the variance and standard deviation for the numbers: 6, 8, 10, 12, 14. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Mean]** x_{bar} = (6 + 8 + 10 + 12 + 14) over 5 = 50 over 5 = 10. [1 Mark]

• **[Step 2: Squared Deviations]** $(6 - 10)^2$ = 16; $(8 - 10)^2$ = 4; $(10 - 10)^2$ = 0; $(12 - 10)^2$ = 4; $(14 - 10)^2$ = 16. Sum = 16 + 4 + 0 + 4 + 16 = 40. [1 Mark]

• **[Step 3: Variance and SD]** Variance sigma^2 = 40 over 5 = 8. Standard deviation σ = √8 = 2 √2 ≈ 2.83. [1 Mark]

• **[Boxed Result: Variance = 8, Standard Deviation = 2 √2 ≈ 2.83]**

6. Find the mean and variance for the frequency distribution:nClass: 0-30, 30-60, 60-90, 90-120, 120-150, 150-180, 180-210nFrequency: 2, 3, 5, 10, 3, 5, 2. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Midpoints x_i & Step Deviations u_i]** Midpoints x_i: 15, 45, 75, 105 (Assumed mean A = 105, h = 30), 135, 165, 195. Step deviations $u_i = (x_i - 105)$ over 30: -3, -2, -1, 0, 1, 2, 3. $f_i u_i$: 2(-3)=-6, 3(-2)=-6, 5(-1)=-5, 10(0)=0, 3(1)=3, 5(2)=10, 2(3)=6. Sum $f_i u_i$ = -17 + 19 = 2. Total N = 30. [1.5 Marks]

• **[Step 2: Mean Calculation]** Mean $x_{bar} = A + h$ ($\Sigma f_i u_i$ over N) = 105 + 30 (2 over 30) = 105 + 2 = 107. [1 Mark]

• **[Step 3: Variance Calculation]** $f_i u_i^2$: 2(9)=18, 3(4)=12, 5(1)=5, 10(0)=0, 3(1)=3, 5(4)=20, 2(9)=18. Sum $f_i u_i^2$ = 76. Variance $sigma^2 = h^2$ [($\Sigma f u^2$ over N) - ($\Sigma f u$ over N)^2] = 900 [76 over 30 - (2 over 30)^2] = 900 [2.5333 - 0.0044] = 900 [2.5289] = 2276. [1.5 Marks]

• **[Boxed Result: Mean = 107, Variance = 2276]**

7. The sum and sum of squares corresponding to length x (in cm) and weight y (in g) of 50 plant products are: Sigma x = 212, Sigma x^2 = 902.8, Sigma y = 261, Sigma y^2 = 1457.6. Which is more variable, length or weight? [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Length x Mean and SD]** Mean x_{bar} = 212 over 50 = 4.24 cm. Variance $sigma_x^2$ = (902.8 over 50) - $(4.24)^2$ = 18.056 - 17.9776 = 0.0784 $\Rightarrow$ $sigma_x$ = $\sqrt{}$(0.0784) = 0.28 cm. C.V.(x) = (0.28 over 4.24) x 100% = 6.60%. [1.5 Marks]

• **[Step 2: Weight y Mean and SD]** Mean y_{bar} = 261 over 50 = 5.22 g. Variance $sigma_y^2$ = (1457.6 over 50) - $(5.22)^2$ = 29.152 - 27.2484 = 1.9036 $\Rightarrow$ $sigma_y$ = $\sqrt{}$(1.9036) $\approx$ 1.38 g. C.V.(y) = (1.38 over 5.22) x 100% = 26.44%. [1.5 Marks]

• **[Step 3: Comparison]** Since C.V.(y) = 26.44% >> C.V.(x) = 6.60%, weight is much more variable than length. [1 Mark]

• **[Boxed Result: Weight is significantly more variable (C.V. 26.4% vs 6.6%)]**